

RIC-Net: A plant disease classification model based on the fusion of Inception and residual structure and embedded attention mechanism

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ABSTRACT

In this paper, we proposed a convolutional neural network based on Inception and residual structure with an embedded modified convolutional block attention module (CBAM), aiming to improve the classification of plant leaf diseases. Corn, potatoes and tomatoes are the most cultivated grains in southern China. The leaves of the three crops are very fragile and sensitive and are susceptible to leaf diseases, such as leaf blight of corn, late blight of potato and mosaic virus of tomato. These diseases cannot be identified at early stages. Therefore, an efficient solution is proposed by deep learning techniques to detect the disease categories of crops, which can effectively prevent the spread of diseases and ensure the normal growth of plants. In this experiment, our model achieved an overall accuracy of 99.55% for the identification of the three diseases of corn, potato and tomato. In addition, we tested the three plants individually. The classification accuracy of our model on corn, potato and tomato was 98.44%, 99.43% and 95.20%, respectively. We have also developed a web-based real-time plant disease classification system and deployed our model. The system had good performance in time and accuracy evaluation metrics. The results of this study showed that our model had fewer parameters, shorter training time, and higher recognition accuracy compared to existing image classification models.

1. Introduction

Plant diseases damage important plant functions by altering the original state of the plant (Shurtleff et al., 2020). There are many types of plant diseases, and more plant diseases occur on the leaves of plants. Corn, potatoes and tomatoes are some of the more abundant crops in southern China. Corn crops have relatively large leaves and the crop height is susceptible to certain foliar diseases such as gray spot, common rust and northern leaf blight (Ouppaphan, 2017). Potato leaves are relatively small and foliar diseases are more pronounced and common diseases include early blight and late blight (Aparajita et al., 2017). Tomato leaves are irregularly shaped, and they vary in size. In addition to early blight and late blight, which are identical to potato, it has a bacterial spot and tomato mosaic virus (Sabrol and Satish, 2016). When these diseases occur, they cause significant damage to the plant leaves and the plant will not be able to photosynthesize, thus losing important nutrients needed for growth and causing great losses to the quality and yield of the crop (Lincy and Rubia, 2021).

In recent years, with the rapid development of image processing technology and machine learning technology, a new method of plant

disease identification has been introduced to solve the problem of low efficiency traditional manual identification. Qin et al. (2009) completed the detection of citrus rots using hyperspectral reflectance imaging, and the overall accuracy of the experiment was 96.2%. Rumpf et al. (2010) accomplished early disease identification of sugar beets using SVM (Support Vector Machine) and hyperspectral techniques. The experimental results showed an accuracy of 97% for healthy and diseased leaves of sugar beet. Johannes et al. (2017) developed a mobile device for real-time detection of plant diseases based on image processing techniques and proposed a novel image processing algorithm based on candidate hot spot detection and statistical inference methods, which provides a new approach for early disease identification. Iqbal et al. (2018) studied various image processing techniques for the detection of citrus plant diseases and preprocessed, segmented, extracted and classified citrus images. The results of the study showed that K-means performed well in disease segmentation, while SVM and NN (Neural Network) performed well in classification.

To address the problem of low recognition rate of complex images by machine learning techniques, the development of deep learning has brought faster speed and higher accuracy for image classification and

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Table 1
Convolutional neural network in plant disease identification related research.

Reference	Network	Dataset	Accuracy
Lu et al. (2017)	CNNs	Rice diseases images	95.48%
Ma et al. (2018)	DCNN	Four cucumber diseases	93.4%
Ferentinos. (2018)	VGG	PlantVillage	99.53%
Too et al. (2019)	DenseNet	PlantVillage	99.75%
Zhang et al. (2018)	GoogLeNet and Cifar10	Corn leaf disease	98.9%, 98.8%
Picon et al. (2019)	ResNet	Five crops and 17 diseases	98%
Kamal et al. (2019)	The reduced MobileNet	PlantVillage	98.34%
Atila et al. (2020)	EfficientNet	PlantVillage	99.97%
Karlekar and Seal. (2020)	SoyNet	Soybean leaf disease	98.14%
Chen et al. (2020c)	DenseNet	Rice diseases	98.63%
Zhao et al. (2021)	CBAM + ResNet	PlantVillage and some real scene pictures	97.59%
B.V and G. (2021)	CNN	PlantVillage	98.93%
Tiwari et al. (2021)	CNN	27 diseases in 6 crops	99.58%

recognition. Notable convolutional neural networks include VGG (Simonyan and Zisserman, 2015), GoogLeNet (Szegedy et al., 2015), ResNet (He et al., 2016a), Inception V4 (Szegedy et al., 2017) and DenseNet (Huang et al., 2017). Although the deeper and wider the network, the more image information can be obtained and the richer the extracted features, more detailed image information is also ignored. Hu et al. (2018) proposed SENet (Squeeze and Excitation Networks). SENet used a new feature recalibration strategy to automatically obtain the importance of each feature channel. Woo et al. (2018) proposed CBAM (Convolution Block Attention Module). CBAM is a feedforward convolutional neural network that combines spatial and channel attention mechanisms. Since it is a lightweight module, it can be integrated into a convolutional neural network structure.

2. Related works

The development of convolutional neural networks provides new solutions to problems in agriculture, which have been investigated by researchers (Table 1). Lu et al. (2017) proposed a method for rice disease identification based on CNNs (Convolutional Neural Networks). Ten diseases were identified using the CNNs model, which achieved a classification accuracy of 95.48%. Ma et al. (2018) proposed a deep convolutional neural network (DCNN) for cucumber disease identification, and the results of the study showed that the DCNN network achieved an accuracy of 93.4% on the test set, providing a reliable tool for cucumber disease identification. Barbedo. (2018a) investigated the impact of multiple factors on deep learning in plant disease classification, providing a wealth of experience in advancing research in this area. Sardogan et al. (2018) used a convolutional neural network model based on the LVQ (Learning Vector Quantization) algorithm for tomato leaf disease classification. The LVQ algorithm is a learnable vector quantization algorithm for training the original image from which the feature vectors from the original image. This study can effectively identify four diseases of tomatoes. Ferentinos. (2018) used multiple deep learning models to detect diseases of plant leaves. The experimental results showed that the VGG network had the highest accuracy of 99.53% in the test set.

Barbedo. (2018b) discussed the application of deep learning and transfer learning in plant disease classification, and the study concluded that the joint use of CNN (Convolutional Neural Network) and transfer learning provided an effective method for plant disease identification. Too et al. (2019) used fine-tuning techniques to evaluate the performance of deep convolutional neural networks in plant disease classification, and the experimental results showed that the DenseNet network

achieved the best results on the PlantVillage dataset with a tested accuracy of 99.75%. Lee et al. (2020) combined deep learning and transfer learning for disease identification. They fine-tuned the PlantVillage dataset on ImageNet using various pre-trained networks. The results showed that the ImageNet pre-trained VGG16 model had better generalization ability in new diseases. The effectiveness of deep learning and transfer learning in plant disease recognition is further demonstrated.

Zhang et al. (2018) proposed an improved GoogLeNet and Cifar10 model for corn leaf disease recognition by adjusting the parameters of the GoogLeNet and Cifar10 models, adding dropout and ReLU functions, and reducing the number of classifiers. The improved GoogLeNet and Cifar10 models achieved an average recognition accuracy of 98.9% and 98.8%, respectively. Picon et al. (2019) proposed a multiplicative suppression method based on ResNet networks. The method can extract disease feature information in greater detail and accuracy and uses contextual information to distinguish similar diseases. The experiments achieved a balance accuracy of 98%. Kamal et al. (2019) embedded the constructed deeply separable convolutional architecture into a MobileNet network and compared SGD, Adam and Nadam optimizers. The experimental results showed that the reduced MobileNet classification accuracy on the PlantVillage dataset reached 98.34%, with better performance of the Adam optimizer and faster network convergence. Atila et al. (2020) used the deep learning architecture EfficientNet for the classification of 39 diseases on the PlantVillage dataset and compared it with other deep learning models. The best average accuracy of 99.97% was achieved. Karlekar and Seal. (2020) proposed the SoyNet network for soybean leaf disease identification. The SoyNet network is divided into two phases. The first stage removes the effect of image background information and extracts the soybean leaf part. The second stage is the disease identification of the extracted soybean leaves using the CNN network. The results of the study showed that the classification accuracy of SoyNet network reached 98.14%.

Chen et al. (2020a) used a VGG network pre-trained on ImageNet for plant disease detection. They improved the final output layer of the VGG network and fed the feature information extracted from the VGG network to the Inception module to obtain the final classification probability. The results showed that the average classification accuracy of the model was 92.00% in the mixed dataset of maize and rice. Chen et al. (2020b) proposed an automatic plant disease detection and classification method based on the GMDH-Logistic model, which uses image processing techniques to analyze the features of the original image, establish the metrics of the prediction model, and feed the selected features to the GMDH-Logistic model to obtain the final results, which can accurately detect diseased plants. Chen et al. (2020c) proposed a model that combines a pre-trained DenseNet network on ImageNet with Inception. Also, they replaced the cross-entropy loss function with the focus loss function. The model achieved an average recognition accuracy of 98.63% for rice diseases.

Zhao et al. (2021) embedded an improved CBAM module into a ResNet network and discussed the effect of the embedding position of CBAM on plant disease classification. The improved model achieved 95.37% classification accuracy for tomato diseases and 97.59% classification accuracy for various diseases. B.V and G. (2021) proposed an effective loss fusion convolutional neural network for plant disease identification, which combined the advantages of both loss functions and showed good results on the PlantVillage dataset with an accuracy of 98.93%. Tiwari et al. (2021) proposed a dense convolutional neural network to identify 27 diseases of six crops and optimized the dense convolutional neural network on a complex and challenging dataset. The network achieved an average cross-validation accuracy of 99.58%.

The above studies have demonstrated the feasibility of convolutional neural networks in the field of plant disease identification. However, convolutional neural networks are weak in recognizing certain diseases due to the specificity of diseased leaf spots. The research needs to solve the problem of identifying specific diseased leaves with convolutional neural networks.

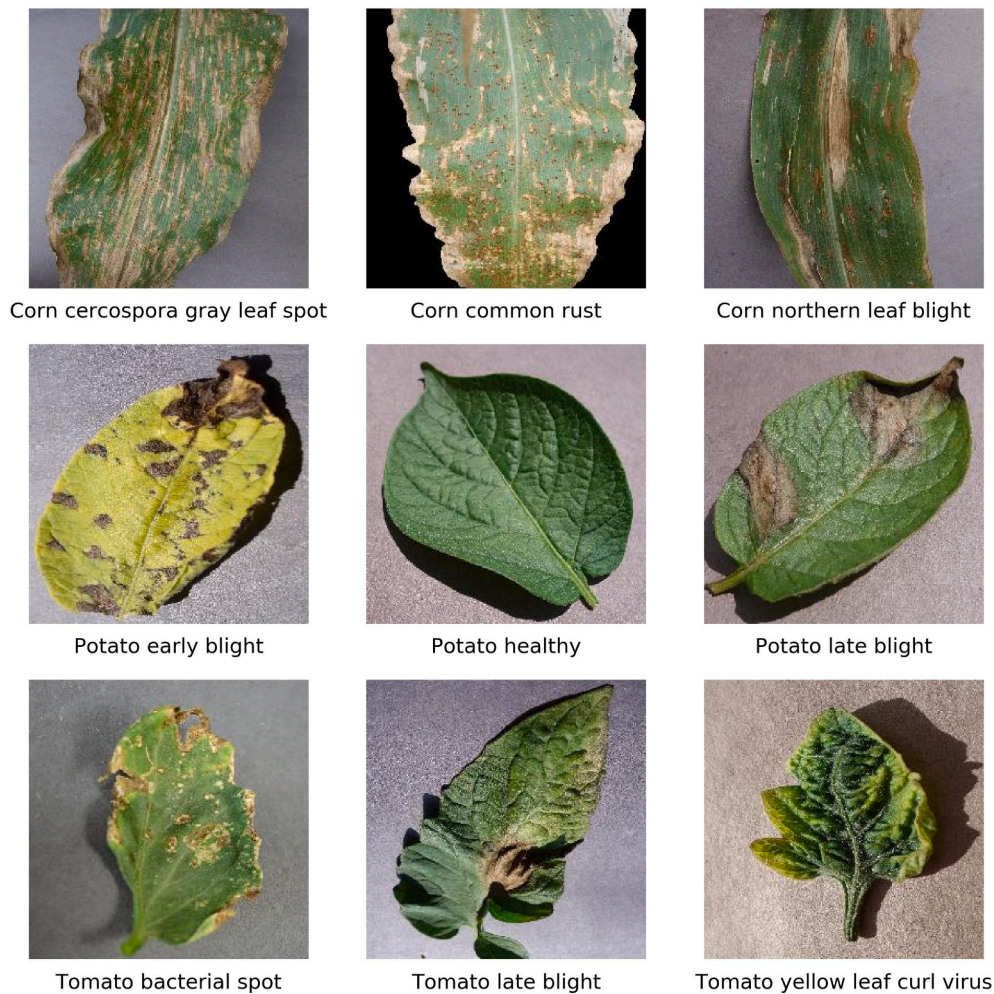


Fig. 1. 9 different plant diseases in the PlantVillage dataset.

Based on the above research and analysis, this paper proposed a method to construct a multi-model fusion network based on convolutional neural networks for plant disease classification using existing deep learning models. The main contributions to this paper are as follows.

(a) Starting from the width and depth of the network, the residuals were merged into Inception as a branch of convolutional layers in the Inception structure, and this fusion module was named RI-Block. The RI-Block module was used to build a convolutional neural network for plant disease recognition, which solved the problem of increasing the computational complexity and number of parameters of traditional convolutional neural networks due to the stacking of convolutional layers. This module was used to build a convolutional neural network for plant disease identification.

(b) Due to the characteristics of large inter-class differences and small intra-class differences of plant disease spots, this paper proposed adding a CBAM module after each RI-Block module to locate the location of disease spots in the image more accurately and improve the recognition accuracy of the model. In addition, to explore the effect of the weighting operation on the CBAM module, the output of the RI-Block module was multiplied once with the output of the CBAM module, and an experimental comparison was performed.

(c) To explore the effect of the channel attention module in the CBAM module on the performance of plant disease classification, the results of the AvgPool and MaxPool layers in the channel attention module were input into two 1D convolutions and the kernel size values of the 1D convolutions were adjusted to qualitatively analyze the effect of

network classification.

(d) A web-based plant disease detection application was designed for this research component. The fully trained RIC-Net model is deployed in the web application, which can accomplish real-time disease identification in a real environment with high accuracy.

3. Dataset and equipment

3.1. Dataset

The dataset used in this paper is from PlantVillage (Hughes and Salathé, 2015), which organizes images according to plant species and disease categories. The plantvillage contains 14 plant species, each with a different plant disease, forming 38 categories. In total, there are 54,305 images, each with R, G, and B channels and an image size of 256×256 . In this study, for three plants that are most abundant in southern China, namely corn, potato, and tomato, we extracted all disease images for these three plants from the PlantVillage dataset to form our experimental dataset (17 categories, 24,164 images). Among them, corn has 4 diseases with 3852 images, potato has 3 diseases with 2152 images, and tomato has 10 diseases with 18,160 images. Fig. 1 depicts the 9 diseases of corn, potato and tomato.

3.2. Image augmentation

The number of samples in a dataset affects the performance of an image classification network. A small number of datasets may lead to

Table 2
Detailed dataset.

Classification	Partitioning of dataset			Data Augmentation	
	Original counts	Train counts	Test counts	Train counts	Test counts
Corn cercospora gray leaf spot	24,164	23,892	272	32,482	5984
Corn common rust	513	497	16	994	352
Corn healthy	1192	1176	16	1688	352
Corn northern leaf blight	1162	1146	16	1658	352
Potato early blight	985	969	16	1481	352
Potato healthy	1000	984	16	1496	352
Potato late blight	152	136	16	576	352
Tomato bacterial spot	1000	984	16	1496	352
Tomato early blight	2127	2111	16	2623	352
Tomato healthy	1591	1575	16	2087	352
Tomato late blight	1909	1893	16	2405	352
Tomato leaf mold	952	936	16	1448	352
Tomato septoria leaf spot	1771	1755	16	2267	352
Tomato spider mites two-spotted	1676	1660	16	2172	352
Tomato target spot	1404	1388	16	1900	352
Tomato mosaic virus	373	357	16	842	352
Tomato yellow leaf curl virus	5357	5341	16	5853	352

poor generalization of complex network models. To enlarge the dataset, this paper performed a batch enhancement process on the data, that is, 32 images were randomly selected from each batch, and the following five enhancement methods were applied to these 32 images in turn. 1) Randomly rotate the images by angles between 20 and 30 degrees. 2) Randomly perform horizontal and vertical offsets between 0 and 0.2 times the height of the images. 3) Perform a cross-cutting transformation on the image pixels. The rule of the transformation is to leave the y-coordinate of all pixels unchanged and the corresponding x-coordinate panned by 0.1 times the x-value. 4) Randomly scale the width and height of the image between 1 - 0.1 times the width and 1 + 0.1 times the width at the same time. 5) Randomly flip the horizontal and vertical orientation of the image. The enhanced dataset has 38,466 images, including

32,482 images in the training set and 5984 images in the test set. Table 2 shows the detailed dataset.

3.3. Equipment

All data in this experiment were run on the following devices. The processor is Intel(R) Core (TM) i5-10400 CPU @ 2.90 GHz, memory 16 GB, graphics card Nvidia GeForce RTX 2060, video memory 6 GB, operating system Windows 10. The software system uses PyCharm2020, python3.7.0, tensorflow-gpu2.0.0 and sklearn library.

4. Methods

The emergence and spread of foliar diseases during the growth and development stages of crops can damage the nutrient absorption capacity of crops, leading to a decrease in crop yield and quality. At this time, deep learning techniques can be used to identify the types of foliar diseases early and accurately and apply the correct methods for disease prevention. To effectively identify disease types, a deep learning network based on multi-model fusion and embedding improved attention was proposed in this paper.

4.1. RI-Net network

Traditional convolutional neural networks improve image classification performance by stacking convolutional layers, which leads to an increase in the number of parameters and computational complexity, and the fixed size of convolutional kernels may lead to the loss of some feature information of plant spots, which affects the recognition rate. To improve the performance of image classification, we built a feature extraction module consisting of three convolutional blocks named RI-Block from the perspective of network width. Two 3×3 convolutional receptive fields of convolutional block 1 in the RI-Block module are equivalent to a 5×5 convolutional receptive field, which can detect relatively large disease spots in plants and obtain the global feature information of diseases. Convolution block 2 contains a 3×3 convolution, which can detect relatively small disease spots in the plant and obtain local feature information of the disease. As shown in Fig. 2, the combination of convolution block 1 and convolution block 2 reduces the loss of partial feature information of plant diseases and improves the recognition rate. In order to solve the problems of gradient disappearance and model optimization difficulty due to too many convolution layers, the convolution block 3 of the RI-Block module adopts the residual structure (He et al., 2016b). The advantage of the residual structure is that it can solve the degradation problem of increasing network depth, because the mapping relationship between the

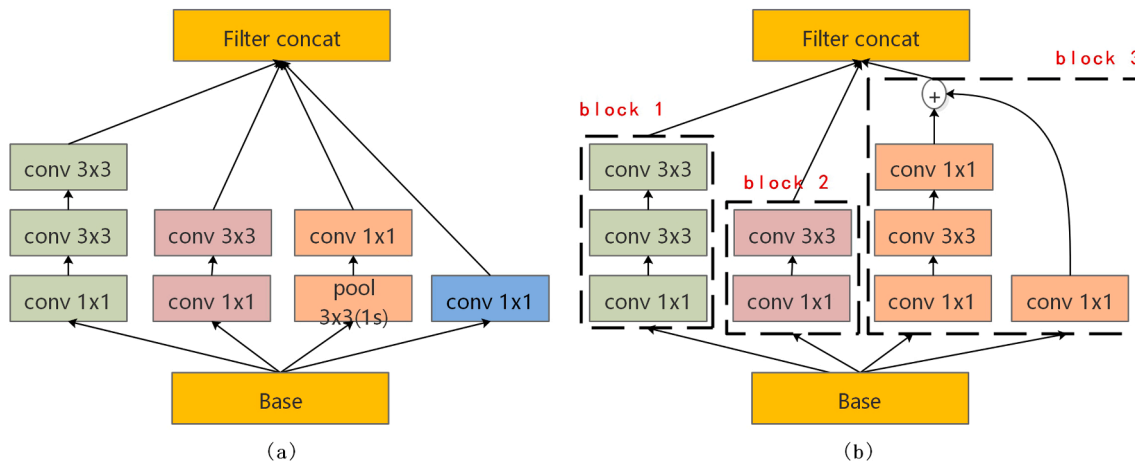


Fig. 2. Inception structure and RI-Block structure: a) Inception structure; b) RI-Block structure.

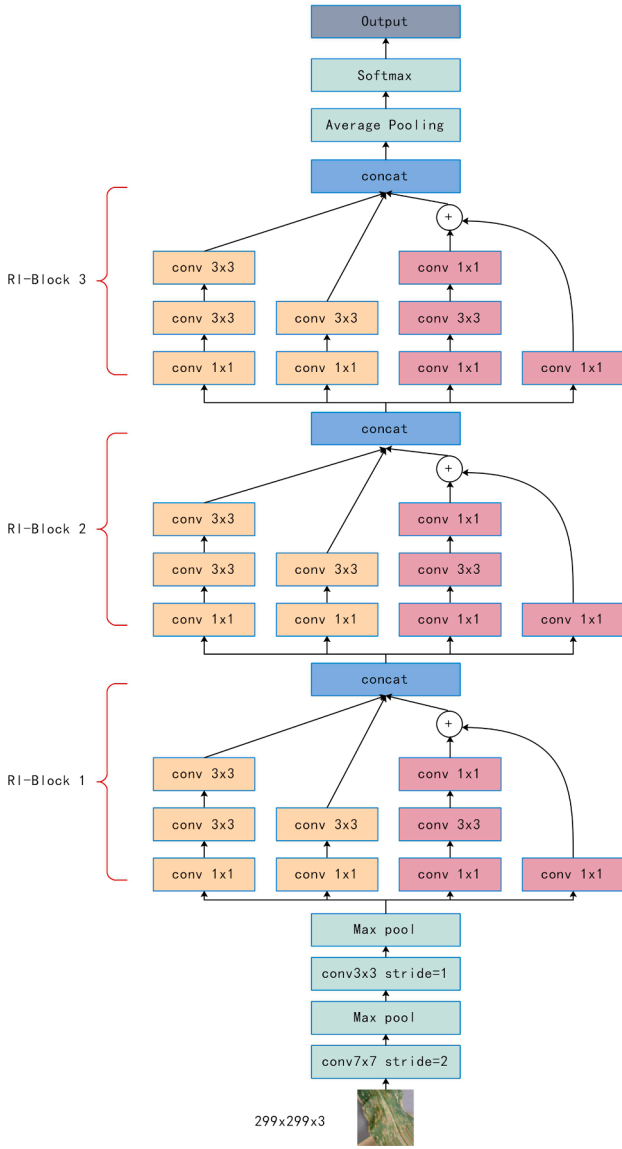


Fig. 3. RI-Net network structure.

convolutional layers of the residual structure is $H(x) = F(x) + x$. As long as $F(x) = 0$, i.e., a small update of some of the weight values can make the network optimal again. To make the RI-Block module better fit the input data, we use 1×1 convolution in all three convolution blocks. The 1×1 convolution allows for downsampling and upsampling operations, reducing computational parameters and flexibly adjusting the output size. Fig. 2 shows the Inception structure and the RI-Block structure.

The network constructed by the Inception structure has too many parameters, large model size and long training time. To solve the problem of the Inception network, we constructed a plant disease classification network with three simplified RI-Block modules named RI-Net. Each RI-Block module in the RI-Net network outputs a different number of channels and obtains twice the number of feature maps than the previous module. The RI-Block1 module outputs 256 feature maps, the RI-Block2 module outputs 512 feature maps, and the RI-Block3 module outputs 1024 feature maps. The feature extraction module consisting of three RI-Blocks can first perform local perception of plant disease features in the images, and then perform higher-level synthesis operations on the obtained local information to obtain the full information. The RI-Net network constructed with the RI-Block module has a wider network structure, which can freely select better features and reduce the complexity of network parameters and computations. Fig. 3

Table 3
RI-Net network parameters.

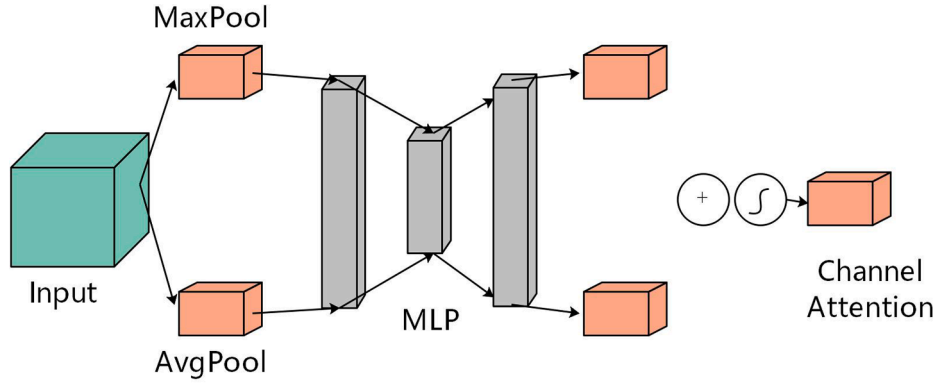
Layer	Kernel	Stride	Output Shape
Input			(None,299,299,3)
Conv2d_1	7×7	2	(None,150,150, 64)
Max pool_1	3×3	2	(None,75,75,64)
Conv2d_2	3×3	1	(None,75,75,128)
Max pool_2	3×3	2	(None, 37,37,128)
RI-Block 1			(None,37,37,256)
RI-Block 2			(None,37,37,512)
RI-Block 3			(None,37,37,1024)
Global average pooling			(None,1024)
SoftMax			(None,17)

shows the network structure of RI-Net. Table 3 shows the relevant parameters of the RI-Net network.

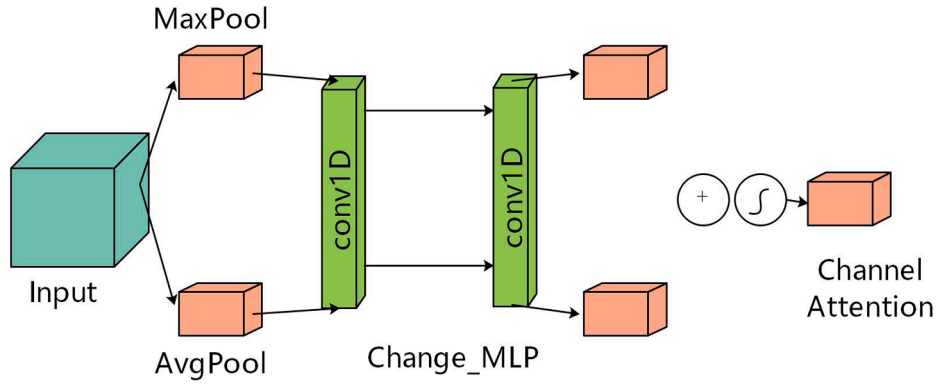
4.2. RIC-Net network

Due to the characteristics of plant disease spots with large inter-class differences and small intra-class differences, the RI-Net network easily causes confusion when recognizing different types of plant disease samples. In order to suppress the influence of redundant information in the input image on the network classification performance, this paper proposed embedding the modified CBAM in the RI-Net network, which can more accurately locate the location of disease spots in the image and improve the recognition accuracy of the model. The proposed modified CBAM is a channel and spatial attention module that can be well integrated with the RI-Net network. The input feature map accepted by the channel attention module contains information in three dimensions. Height (H), Width (W) and Channel (C). The feature map (F) of a given $H \times W \times C$ shape will undergo average pooling and maximum pooling to obtain two different feature descriptions, F_{avg}^C and F_{max}^C , which are then input to the Change_MLP (Change Multilayer Perceptron) layer. In order to avoid the corruption of plant spot feature information by the dimensionality reduction operation of the original MLP layer, we replaced the MLP layer with two 1D convolutions. 1D convolution does not involve the dimensionality reduction operation; it is a featured information detector. The size of the 1D convolutional kernel size is analyzed qualitatively through experiments so that it can better detect disease spot information of different sizes in F_{avg}^C and F_{max}^C when it takes a certain value. Next, the features of F_{avg}^C and F_{max}^C are combined using the join operation to generate the channel attention map $M_C(F) \in R^{1 \times 1 \times C}$, and $M_C(F)$ is input to the spatial attention module to generate the spatial attention map $M_S(F) \in R^{H \times W}$. In the experiment, we noted that $M_S(F)$ was the score of diseased and non-diseased spots in the original feature map (F), i.e., the proportion of diseased and non-diseased spots in the image. If $M_S(F)$ is directly output to the next convolution module, the convolution calculation will be affected by the maximum and minimum values, so we added a weighting operation to highlight the importance of lesions. The weighting operation is a channel-by-channel weighting of the proportion of $M_S(F)$ output to the original image by multiplication. By the weighting operation, the information about the disease spots in the original image is further magnified, while the information about the non-disease features is suppressed. The feature maps obtained by the weighting operation are used to improve the model's ability to identify plant diseases and are used as input data for the next layer. Fig. 4 shows the modified channel attention mechanism module. Fig. 5 shows the modified convolutional attention module.

Due to the specificity of plant spots and the poor adaptability of the original CBAM to the RI-Net network, in order to improve the recognition rate of the RI-Net network for spots of different shapes and sizes, we established RIC-Net, a plant disease classification network, based on the RI-Net network and Change_CBAM module. The input to the RIC-Net network is a plant disease classification network containing R, G and



(a) Original Channel Attention Module



(b) Change Channel Attention Module

Fig. 4. a) Original Channel Attention Module; b) Modified Channel Attention Module, replacing the operations in Shared MLP with two one-dimensional convolutions.

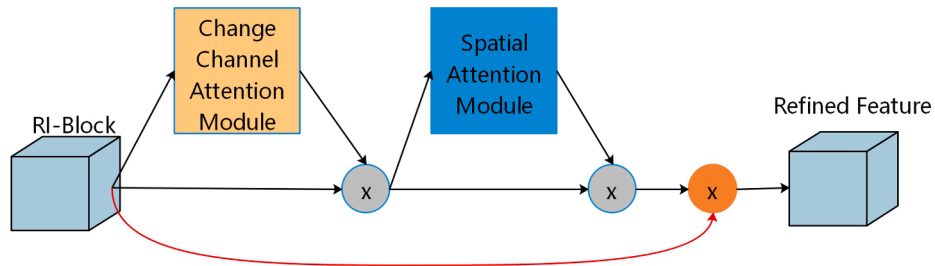


Fig. 5. Change_CBAM, two improvements to CBAM: 1) Change Channel Attention Module is to modify the Shared MLP; 2) Add a weighting operation.

B channels of plant disease images. Three RI-Block modules are used in the middle layer, and the output data of each RI-Block module will be passed through the Change_CBAM module for more accurate extraction of image features. Finally, the SoftMax layer is used to calculate the recognition probability for each type of disease. The network model with hybrid attention modules embedded in the channel domain and spatial domain can identify plant diseases more finely and filter the feature information of non-disease parts to improve the accuracy of model identification. Fig. 6 shows the RIC-Net network.

4.3. Evaluation metrics

For the training results, we use the accuracy rate, precision rate and recall rate to evaluate the model. Accuracy is the proportion of the number of diseases predicted to be correct in the number of all diseases,

as shown in (1). Precision is the number of diseases predicted to be correct for a certain type of disease, as shown in (2). Recall is the proportion of the number of diseases that are predicted to be correct for a certain type of disease, as shown in (3).

$$accuracy = \frac{TP + FN}{TP + TN + FP + FN} \quad (1)$$

$$precision = \frac{TP}{TP + FP} \quad (2)$$

$$recall = \frac{TP}{TP + FN} \quad (3)$$

where TP is the true positives, TN is the true negatives, FP is the false positives and FN is the false negatives.

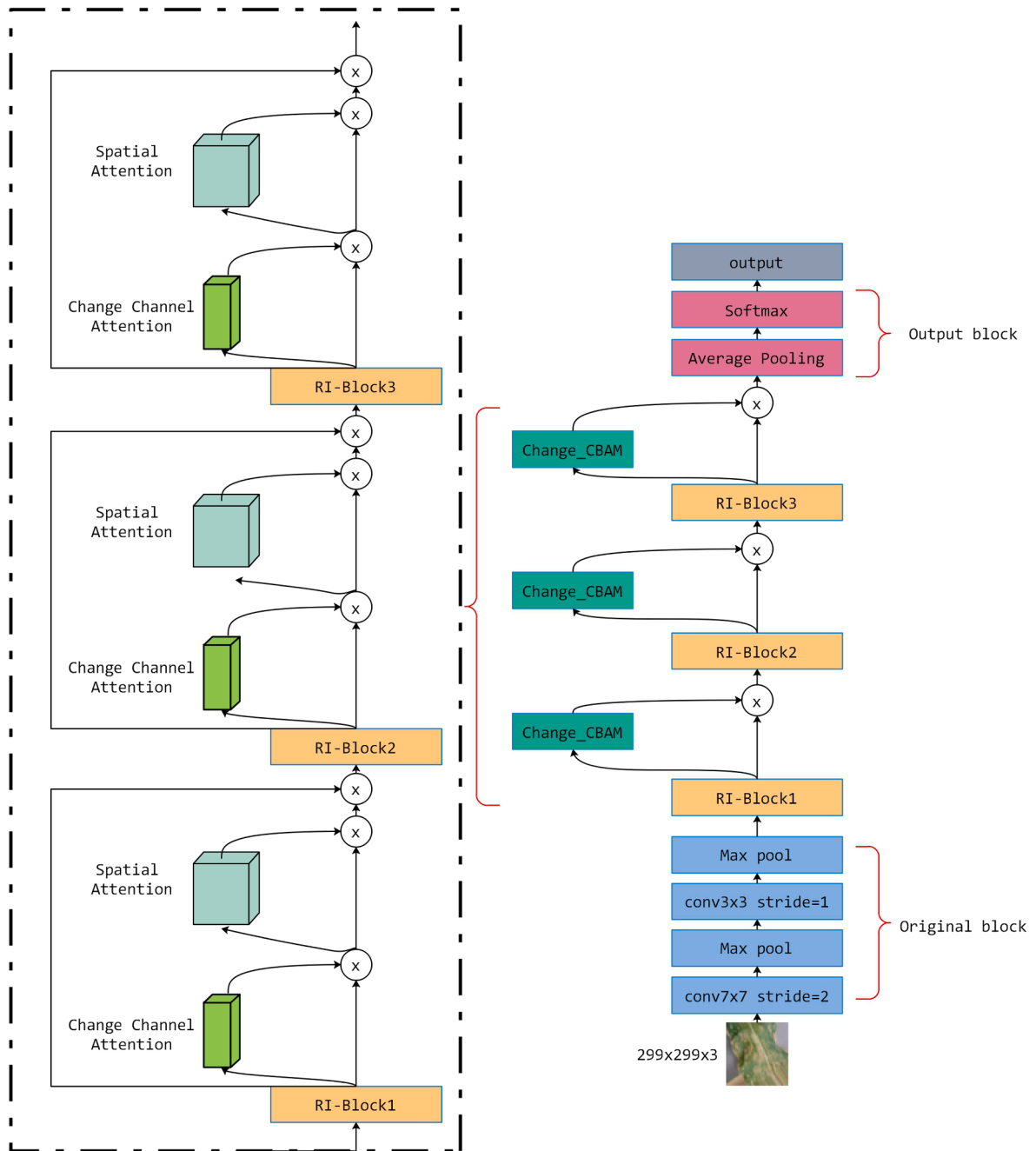


Fig. 6. RIC-Net network.

5. Experiments and results

The main objective of this study is to investigate the classification performance of a deep learning network based on Inception and residual structure fusion for multiple and single plant diseases and to compare it with InceptionV3, Xception, ResNet50 and Inception-ResNetV2. All

experiments were conducted on an enhanced dataset. There are 17 plant diseases with 32,482 images in the training set and 5,984 images in the test set. To maximize the performance of the experimental equipment and to overcome the problem of large amounts of data, we set the batch size value to 16. During training, the training data was divided into 2030 batches of data and the batches were iterated. As the number of epochs increases, the number of weight update iterations in the neural network increases, and the curve gradually enters the optimized fitting state from the initial unfitting state, and finally enters the overfitting state at the 20th epoch. We set the epoch at 20, and the training will automatically save the optimal model. There are 17 diseases in our dataset, which are multi-classification problems. We use the cross-entropy loss function. For the multi-classification problem, it is simple to derive and is only related to the probability of the correct category. Since the Adam optimizer is good at handling sparse gradients and non-stationary targets,

Table 4
Training parameters.

Parameter	Value
Optimizer	Adam
Loss function	Cross-entropy
Batch size	16
Epochs	20
Learning rate	0.001

training loss curves of 7 trained models

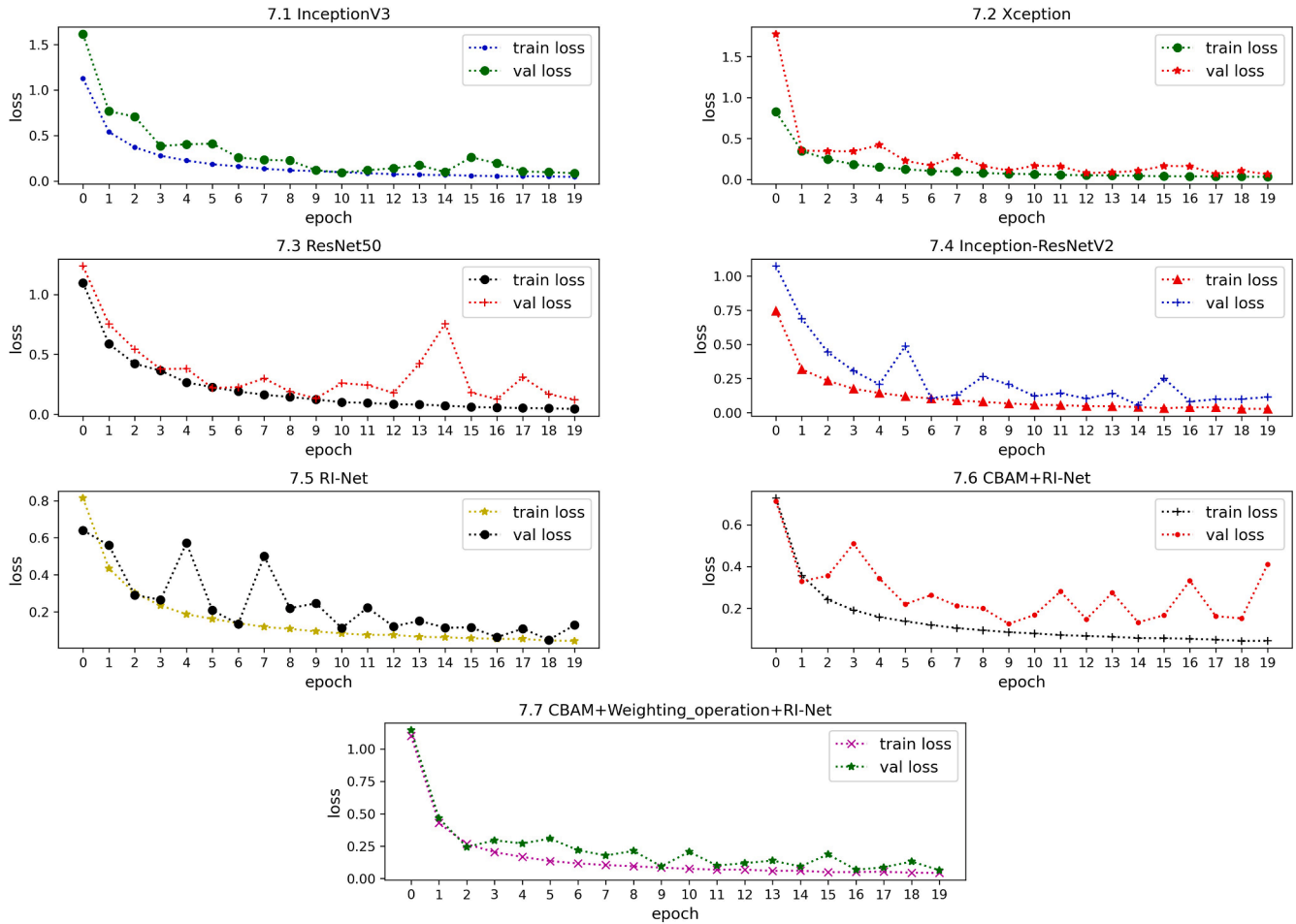


Fig. 7. Training loss curves of 7 trained models.

has small memory requirements, and is suitable for large datasets, we used the Adam optimizer for our model training with the learning rate set to 0.001 to accelerate network convergence. The specific training parameters are shown in Table 4.

5.1. Classification results of multiple plant diseases

5.1.1. RI-Net, CBAM and weighting operations

To investigate the effect of CBAM and weighting operations on the classification performance of the model, this paper attempts to analyze the effect of embedding only CBAM and both CBAM and weighting operations in the RI-Net network on the recognition rate of multiple plant diseases. The experimental results showed that for multiple plant disease classification, the accuracy of the RI-Net network on the test set was 95.74%. When only the CBAM module was embedded in the RI-Net model, the integration of the RI-Net network with the CBAM module was poor, probably due to the specificity of plant disease spots, and the classification accuracy of the test set was reduced to 94.59%. However, when we embedded both CBAM and weighting operations, the classification accuracy of the test set reached the best value of 97.96%. In addition, we trained and compared other image classification networks. Among them, InceptionV3, Xception, ResNet50 and Inception-ResNetV2 achieved 97.33%, 97.63%, 96.16% and 95.94% accuracy for the test set, respectively. Comparing the training loss curves of the seven models, we found that the network fitting ability of RI-Net was poor in the first 10 epochs (Fig. 7.5). As the network was continuously trained, the fitting ability of the model gradually increased. By observing the training loss

curve of the CBAM + RI-Net network (Fig. 7.6), we found that the RI-Net network was poorly adapted to the CBAM module. When the epoch was 9, the error rate of the validation set of the model reached the lowest, and then the model entered the overfitting state. However, when we embed CBAM and weighting operations in the RI-Net network, we observed that Fig. 7.7 showed that the model became better fitted. In addition, we compared the techniques reported in the PlantVillage dataset. Kamal et al. (2019) proposed a deep separable convolutional model based on MobileNet. The modified MobileNet model achieved an accuracy of 97.65% for the PlantVillage dataset. Chohan et al. (2020) proposed a deep-learning-based plant disease detector using the feature extraction function of CNN. The accuracy of their proposed model tested on the PlantVillage dataset reached 98.30%. From the technical report of the PlantVillage dataset, our model outperformed Modified MobileNet (Kamal et al., 2019), but was significantly lower than the proposed model (Chohan et al., 2020). We read both reports carefully and analyzed that the reason for the difference in test accuracy could be the difference in the number of disease images. Although the dataset was all from PlantVillage, we extracted data from PlantVillage for three plants, corn, potato and tomato, and they used the whole PlantVillage dataset. Our study was highly specific, so there were biases between experimental results. The results showed that the classification performance of various plant diseases was better when both CBAM and weighting operations were embedded into the RI-Net network in this paper. The reason was that the CBAM had a better ability to select plant disease features and the weighting operation had a better ability to filter redundant information about plant diseases. The detailed training

Table 5

Data results of different model training and testing.

Model	Training accuracy%	Training loss	Testing accuracy%	Testing loss
InceptionV3	98.38	0.0463	97.33	0.0900
Xception	98.85	0.0322	97.63	0.0633
ResNet50	98.48	0.0448	96.16	0.1200
Inception-ResNetV2	98.60	0.0413	95.94	0.1116
RI-Net	98.48	0.0430	95.74	0.1263
CBAM + RI-Net	98.42	0.0447	94.59	0.1500
CBAM + Weighting operation + RI-Net	98.43	0.0432	97.96	0.0675
Modified MobileNet (Kamal et al., 2019)			97.65	
Proposed model (Chohan et al., 2020)	99		98.3	

results are shown in Table 5.

5.1.2. RIC-Net network

To explore the effect of the channel attention module in the CBAM module on plant disease classification performance, we replaced the MLP layer in the channel attention module with two one-dimensional convolutions to accurately obtain the feature information of disease points. In this paper, we qualitatively analyzed the disease classification effect of the RIC-Net network by adjusting the kernel size values of the one-dimensional convolution. The experimental results showed that when the kernel size value was 1, the classification accuracy of the test set was 97.11%. When the kernel size was set to 2, the classification accuracy of the test set was improved and reached 98.09%. When the kernel size value was 3, the test set reached the best value of 99.55% in the classification of plant diseases. However, when we continued to increase the kernel size value, the classification accuracy of the test set would be lower than when the kernel size value was 3. The accuracy of the test set was 98.01%, 96.84% and 97.34% when the kernel size values were 4, 5 and 6, respectively. We analyzed the reason for the poor performance of the model when the kernel size value exceeded 3. This may be due to the different sizes of plant spots, with large differences between clusters and small differences within clusters. With the increasing kernel size, the model paid attention to not only disease information but also most of the non-disease information during the training process. In the final classification of the model, the non-disease information became a kind of redundant information that interfered with disease identification, leading to a decrease in the classification accuracy of the test set. To better explain the effect of different kernel sizes on network classification performance, we used Grad-CAM (Gradient Weighted Class Activation Mapping) for visualization. Grad-CAM (Selvaraju et al., 2017) can show the regions that the model pays attention to after the channel attention module. In this experiment, we used Grad-CAM to output some plant disease pictures after the channel attention module of Change_CBAM. Some of the heat maps are shown in Fig. 8. It can be seen that the model accurately locates the location of plant diseases on the generated heat maps when the kernel size value is 3. The results of the study showed that the modified channel attention module in this paper had better results for plant disease classification. By adjusting the kernel size, we found that the model had the best classification performance when the kernel size value was 3, because the model had better filtering ability for plant non-disease information and better nonlinear selection ability for disease information. The detailed training results are shown in Table 6.

5.1.3. Comparison of model size, training time and inference time

We compared all models together in terms of model size, training time and inference time. The total training time is the time it takes for the model to start training to finish, and the time per epoch is the total training time divided by the total epoch. Inference time is the time

required for the model to predict each image. The experimental results showed that the RI-Net network performed best in terms of model size and training time, with a model size of 19 MB, training time of 113.53 min, and inference time of 0.005 s. When we introduced the original CBAM module and weighting operation into the RI-Net network, the model size, training time, and inference time increased, with a model size of 21.1 MB, training time was 153.25 min, and inference time was 0.006 s. When we modified the channel attention module in the CBAM module, the RIC-Net network size was reduced to 19.1 MB, and the accuracy of the model was greatly improved, although the training time increased to 160.17 min. In addition, we also compared other image classification networks. Combined with Table 7, we saw that our model outperformed InceptionV3, Xception, ResNet50, and Inception-ResNetV2 in terms of model size, training time, and inference time. The study showed that our model had smaller size, shorter training time, and higher accuracy. Table 7 shows the model size, training time, and inference time.

5.1.4. The impact of class weights and adaptive learning rate

Considering the imbalance in the number of samples in the dataset and the effect of the fixed learning rate on the recognition ability of the model, we re-evaluated our method using a correlation approach. To cope with the balance problem in the dataset, we used the class weighting method. Class weighting is a direct modification of the loss function by penalizing classes with different weights. For the problem of fixed learning rate, we used adaptive learning rate. The initial learning rate was set at 0.001. When an epoch passed and the model performance did not improve, the learning rate decreased. The learning rate is reduced in the form of $lr = 0.3 * lr$. To obtain a well-performing neural network, we used an early stopping approach. During training, when the validation loss did not decrease for 3 consecutive epochs, the loss was considered to be no longer decreasing and the model stopped training. The experimental results showed that the accuracy of all models on the test set was improved when using class weights and adaptive learning rate. As shown in Appendix Table A.1, the test accuracies of RI-Net and CBAM + RI-Net were 98.91% and 99.01%, respectively. Although the test accuracies of these two models were lower than those of the 99.08% of InceptionV3 and 99.08% of Inception-ResNetV2, they outperformed the 98.71% of Xception and 98.41% of ResNet50. The CBAM + Weight operation + RI-Net model on the test set with the best performance, with a test accuracy of 99.36%. This result showed that the weighting operation can effectively improve the classification ability of the model. The RIC-Net model had a test accuracy of 99.33%. Although it was lower than before, it was still higher than other models. The results showed that the weighting operation and the modified channel attention module proposed in this paper had better results for plant disease classification. Appendix Fig. A.1 shows the training loss curve of the trained model. Appendix Fig. A.2 shows the learning rate decrease curve of the trained model.

5.2. Classification results of individual plant diseases

To validate the effectiveness of our model in classifying individual plant diseases, this paper used the RIC-Net network to identify disease images of corn, potato and tomato. The experimental results showed that the RIC-Net model achieved an overall test accuracy of 98.44% for corn (98.06% for corn in the optimized DenseNet network proposed by Waheed et al. (2020)). The model was the best at classifying corn common rust and health with 100% accuracy. Our model was less accurate in classifying corn gray leaf spot and northern leaf blight with 97.7% and 96.1% accuracy, respectively. Although the accuracy of our model for northern leaf blight was lower than that of the optimized DenseNet network at 97.0%, our model outperformed the optimized DenseNet network in overall recognition. The reason for our analysis may be that the disease characteristics of gray leaf spot and northern leaf blight did not differ much and were less numerous, so the accuracy was

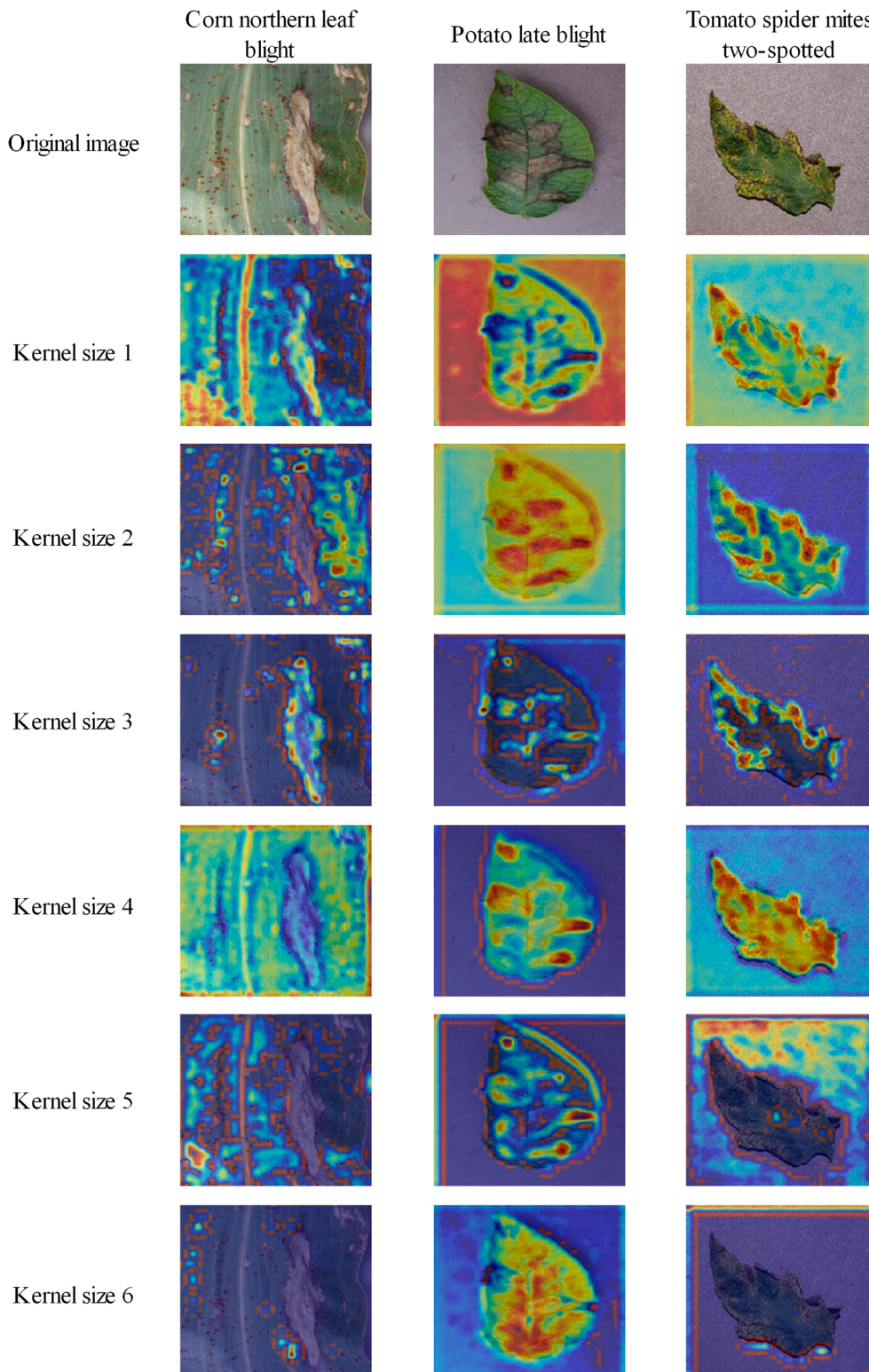


Fig. 8. The heat map of the output.

lower. The overall classification accuracy of the RIC-Net network for potato was 99.43%. The network had the highest recognition rate for early blight and healthy potato with 100% accuracy. The accuracy of the late blight of potato was 98.3%. Our analysis revealed that the network predicted some images of late blight as early blight. This may be due to the fact that the spot characteristics of late blight were similar to those of early blight, so the network judged late blight as early blight. The overall

test accuracy of the RIC-Net network for tomatoes was 95.20%. The network performed best for tomato health, late blight and two-spotted spider mite with a recall of 100%. However, the network did not classify tomato bacterial spot, early blight and mosaic virus well, with recall rates of 79.0%, 90.1% and 92.9%, respectively. We observed the confusion matrix for tomatoes (Fig. 10) and found that the network predicted 73 images of bacterial spot as leaf mold, 26 images of early

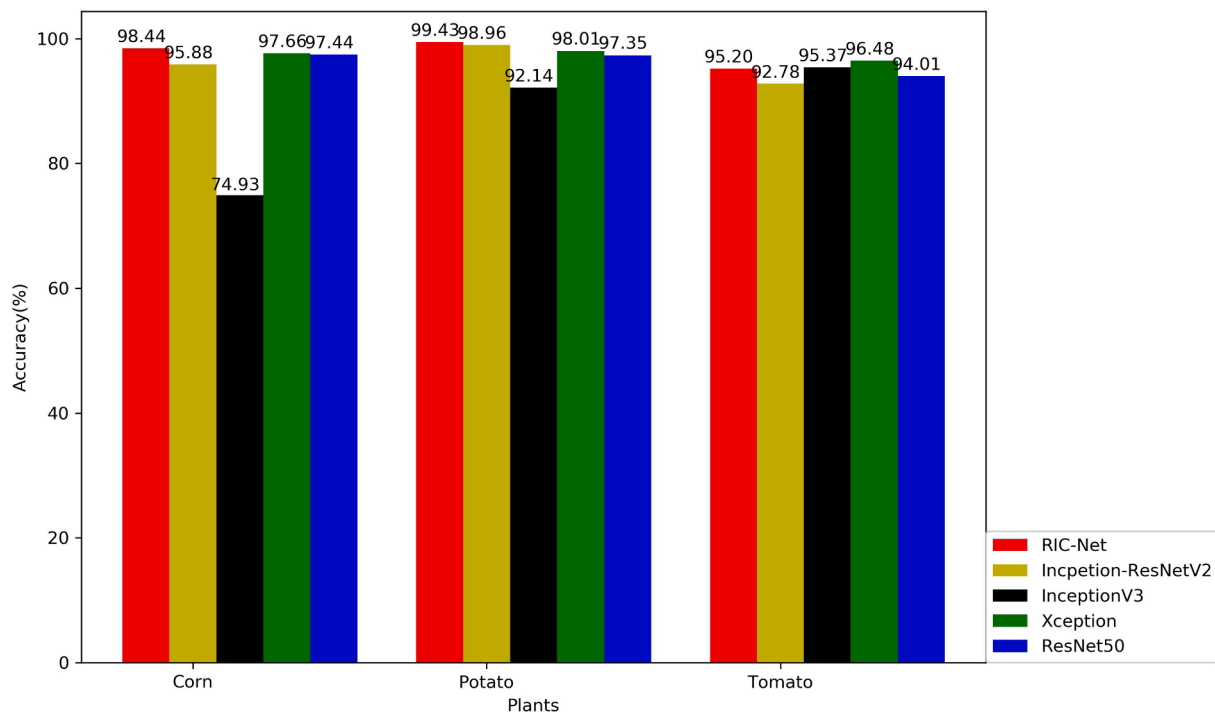


Fig. 9. Test results of networks on corn, potato and tomato.

Table 6

Experimental results of different kernel size values on the train set and test set.

Kernel size value	Training accuracy%	Training loss	Testing accuracy%	Testing loss
1	98.99	0.0295	97.11	0.1041
2	99.09	0.0277	98.09	0.0634
3	99.22	0.0277	99.55	0.0175
4	99.25	0.0237	98.01	0.0507
5	99.23	0.0220	96.84	0.0859
6	99.25	0.0223	97.34	0.0754

Table 7

Data results of model size, training time and inference time.

Model	Size (MB)	Time per epoch (min)	Total training time(min)	Inference time (seconds/per picture)
InceptionV3	250	11.60	232.02	0.015
Xception	239	19.23	384.51	0.011
ResNet50	270	10.11	202.17	0.011
Inception-ResNetV2	624	28.18	563.59	0.039
RI-Net	19	5.68	113.53	0.005
CBAM + RI-Net	21.1	7.48	149.67	0.006
CBAM + Weight_operation + RI-Net	21.1	7.66	153.25	0.006
RIC-Net	19.1	8.01	160.17	0.006

blight as late blight, and 25 images of mosaic virus as Septoria leaf spot. After analyzing the dataset, we found that the reason may be that the differences between plant diseases were small and the network did not obtain enough information about the features of similar spots, so the recognition rate was poor. The results showed that the RIC-Net network had the best recognition for corn and potato compared to Inception-ResNetV2, InceptionV3, Xception and ResNet50. Although the recognition accuracy of the RIC-Net network for tomato was 95.20%, which had some error compared with 95.37% of InceptionV3 and 96.48% of Xception, the RIC-Net network was better than Inception-ResNetV2 and

ResNet50, and the error was within the acceptable range. Fig. 9 shows the test results of different networks on corn, potato, and tomatoes. Tables 8, 9 and 10 show the detailed test results for corn, potato and tomato diseases, respectively.

5.3. Web-based real-time plant disease identification application

With the rapid development of agricultural computers, it is necessary to build an application that can detect plant diseases in real time. In this paper, we proposed a plant disease classification model based on Inception and residual structure fusion, and designed and developed a web-based plant disease detection application, which can detect multiple plant disease types with high recognition accuracy.

There were two important steps in the design and development of a web-based real-time disease identification system. The first step was to send a large amount and multiple types of disease data to the established plant disease classification network RIC-Net for adequate training. The data contained both healthy and diseased leaves. Then, a training model with the best test results was selected, which achieved 99.55% accuracy for the detection of various plant diseases. The second step was to deploy the best model into a web-based program. The program allowed real-time disease detection on computers and cell phones. Data acquisition equipment was used to collect plant disease data from real scenarios. The expected resolution of the provided images is 224×224 . The percentage of diseased parts in the leaves varies from plant to plant due to their own leaf shape. For a large number of samples collected for testing data, they can be transferred to a computer for real-time testing, and for a small number of samples collected for testing data, they can be tested in real-time on a cell phone. The computer terminal can handle multiple samples simultaneously. The detection process of the whole network system is shown in Fig. 11. The main screen of the web system shows three types of result data, which are the prediction category of the image, the accuracy rate and the time spent. Among them, the model is loaded when the system is used for the first time, so the detection time of the first image includes the model loading time, image upload time, processing time and result return time. The detection time for each subsequent image does not include the model loading time. Fig. 12 and

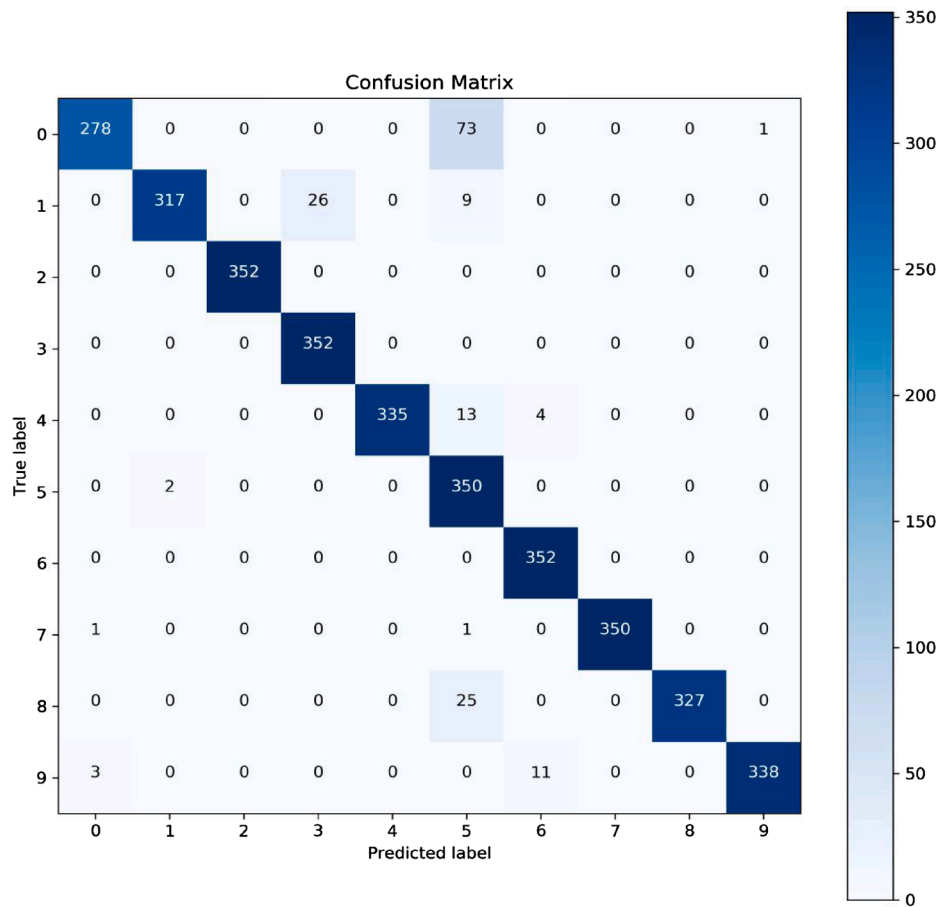


Fig. 10. Confusion matrix of tomato diseases.

Table 8

Detailed results of corn diseases.

Model	Index	Cercospora gray leaf spot	Common rust	Healthy	Northern leaf blight
InceptionV3	precision	0.510	0.975	1.000	0.940
	recall	0.957	0.145	1.000	0.895
Xception	precision	0.914	1.000	1.000	1.000
	recall	1.000	1.000	0.992	0.915
ResNet50	precision	0.954	0.989	1.000	0.955
	recall	0.943	1.000	1.000	0.955
Inception-ResNetV2	precision	0.984	1.000	1.000	0.868
	recall	0.850	1.000	1.000	0.986
Optimized DenseNet (Waheed et al., 2020)	precision	0.920	1.000	1.000	0.970
	recall	0.940	1.000	1.000	0.960
RIC-Net	precision	0.977	1.000	1.000	0.961
	recall	0.960	1.000	1.000	0.977

Fig. 13 show the results of the web-based real-time disease detection data on cell phones and computers, respectively. The experimental results showed that the web-based system can accomplish real-time detection of plant diseases with high recognition accuracy.

6. Discussion

In this paper, we proposed a new image feature extraction structure, i.e., integrating the residual structure into the Inception module. We used it to build a plant disease classification network. Meanwhile, to explore the effect of the attention mechanism on the classification

Table 9

Detailed results of potato diseases.

Model	Index	Early blight	Healthy	Late blight
InceptionV3	precision	1.000	0.989	0.815
	recall	0.776	1.000	0.989
Xception	precision	1.000	0.997	0.946
	recall	0.943	1.000	0.997
ResNet50	precision	0.941	0.986	0.997
	recall	1.000	0.997	0.923
Inception-ResNetV2	precision	1.000	0.972	0.997
	recall	0.997	1.000	0.972
RIC-Net	precision	1.000	1.000	0.983
	recall	0.983	1.000	1.000

performance of our model, we made two improvements to CBAM, i.e., weighting operation and modification of the channel attention module. The experimental results showed that our model improved the classification performance of plant diseases, especially when the modified CBAM was embedded into the model. The modified CBAM can greatly improve the accuracy of the test.

The experimental results on a dataset with a mixture of multiple plants and multiple diseases are shown in Table 11. Kamal et al. (2019) proposed a deep separable convolutional model based on MobileNet. Their Modified MobileNet model was tested with an accuracy of 97.65% on the PlantVillage dataset. Chohan et al. (2020) proposed a deep-learning-based plant disease detector by exploiting the feature extraction function of CNN. Their proposed model achieved a tested accuracy of 98.3% on the PlantVillage dataset. However, our model achieved the best classification results with 99.55% accuracy. The experimental results on a dataset of multiple diseases of one plant are shown in Table 12.

Table 10
Detailed results of tomato diseases.

		InceptionV3	Xception	ResNet50	Inception-ResNetV2	RIC-Net
Bacterial spot	precision	1.000	0.967	0.948	0.997	0.986
	recall	0.952	0.921	0.991	0.869	0.790
Early blight	precision	0.890	0.978	0.926	0.996	0.994
	recall	0.901	0.881	0.787	0.631	0.901
Healthy	precision	0.994	1.000	0.964	1.000	1.000
	recall	1.000	1.000	1.000	0.997	1.000
Late blight	precision	0.859	0.922	0.830	0.780	0.931
	recall	1.000	1.000	1.000	0.977	1.000
Leaf mold	precision	1.000	1.000	1.000	1.000	1.000
	recall	0.960	0.992	0.943	0.972	0.982
Septoria leaf spot	precision	0.953	0.961	0.988	0.784	0.743
	recall	0.813	0.974	0.906	0.980	0.994
Two-spotted spider mite	precision	0.994	0.949	0.864	0.990	0.959
	recall	0.915	0.943	0.955	0.872	1.000
Target spot	precision	0.891	0.926	0.934	0.880	1.000
	recall	0.997	0.989	0.926	1.000	0.994
Mosaic virus	precision	0.997	0.997	0.997	0.983	1.000
	recall	1.000	1.000	0.994	1.000	0.929
Yellow leaf curl virus	precision	0.989	0.957	0.988	0.991	0.997
	recall	1.000	0.949	0.898	0.980	0.960

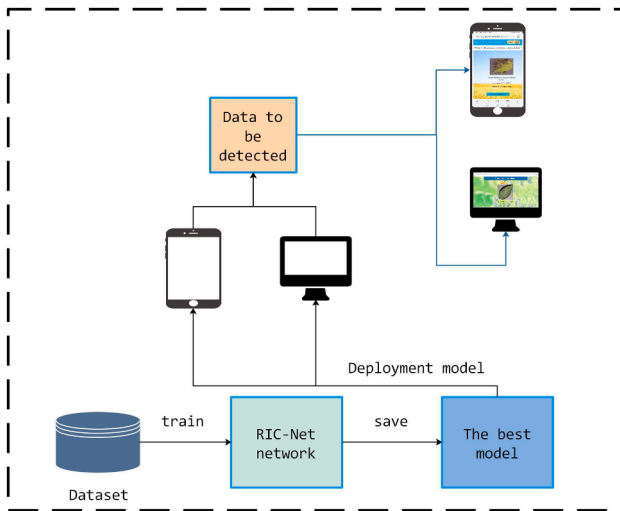


Fig. 11. Process of real-time disease detection system based on web.

In the corn test, the optimized DenseNet network had an accuracy of 98.06% (Waheed et al., 2020). In tomato disease identification, the average test accuracy of the model was 91.2% (Agarwal et al., 2020). The testing accuracy of our model was 98.44%, 99.43% and 95.20% for corn, potato and tomato, respectively. Comparing the techniques reported in the PlantVillage dataset, our model had high accuracy, both for the identification of multiple plant diseases and for the identification of single plant diseases.

In this study, although the dataset had a large number of plants and a rich set of disease features, the images were taken under controlled conditions and only one disease was present in each image. This is thought to be a sign that the model performs well in the laboratory, but has more limitations in practical applications. This is reflected in the potentially high error rate of images taken in the field. We were not able to test the performance of the proposed network on images taken in the field because we did not find suitable data on plant diseases in the field due to the limitations of real scenarios and some uncertainties. Although the proposed model was not used in real scenarios, the results generated by the model provide a wealth of information that can be further explored in future developments.

In this section, we discussed the improvement process of our model, the comparison of experimental results, and issues related to the dataset.

Compared to the already reported techniques, we had two main drawbacks. On the one hand, we studied few disease categories, with three diseases, namely corn, potato and tomato. While the crops were complex and diverse, we needed to extend the crop categories to make our model have better generalization ability. On the other hand, the dataset used in the experiments was clear, weak against interference, and had only one disease on each image. In contrast, in real agricultural plantations, the leaves of crops were complex, so our model had more limitations in practical applications. Although there were shortcomings in our experiments, there were many advantages of our model, whether the experimental results were mixed for multiple diseases or for a single disease, our model had high recognition accuracy compared with existing techniques, and our model was more lightweight, with fewer parameters and shorter training and inference times.

7. Conclusion

In this paper, a convolutional neural network for the classification of leaf diseases of corn, potato and tomato was proposed. The classification network based on the RI-Block module can reduce the training parameters, shorten the training time and improve the training stability by integrating the idea of residuals into the Inception module. It can not only extract the feature information of plant disease spots at multiple scales, but also solve the degradation problem of deep networks. Meanwhile, we modified CBAM in two ways. one was to replace the MLP layer with two one-dimensional convolutions. One-dimensional convolution is a feature detector that not only avoids the corruption of image feature information by the dimensionality reduction operation, but also captures the information interaction across channels. The recognition of leaf disease information by the network is optimized by adjusting the kernel size of the 1D convolution. Experiments showed that the recognition rate of the model was the best when the kernel size value was 3. Next, the recalibration of the feature information was accomplished using weighting operations. Adding the modified CBAM network improved the recognition rate of the model for different diseases. To validate the performance of the model, we extracted three plant images, corn, potato and tomato, from the PlantVillage dataset. For the extracted data, we divided the training and test sets and performed data enhancement. Compared with InceptionV3, Xception, ResNet50 and Inception-ResNetV2, our model had higher running efficiency and recognition accuracy with 99.55% accuracy and a model size of 19.1 MB. In addition, we also recognized these three plants separately, and our model had the recognition accuracy of corn, potato and tomato. The recognition accuracy of our model was 98.44%, 99.43% and 95.20% for

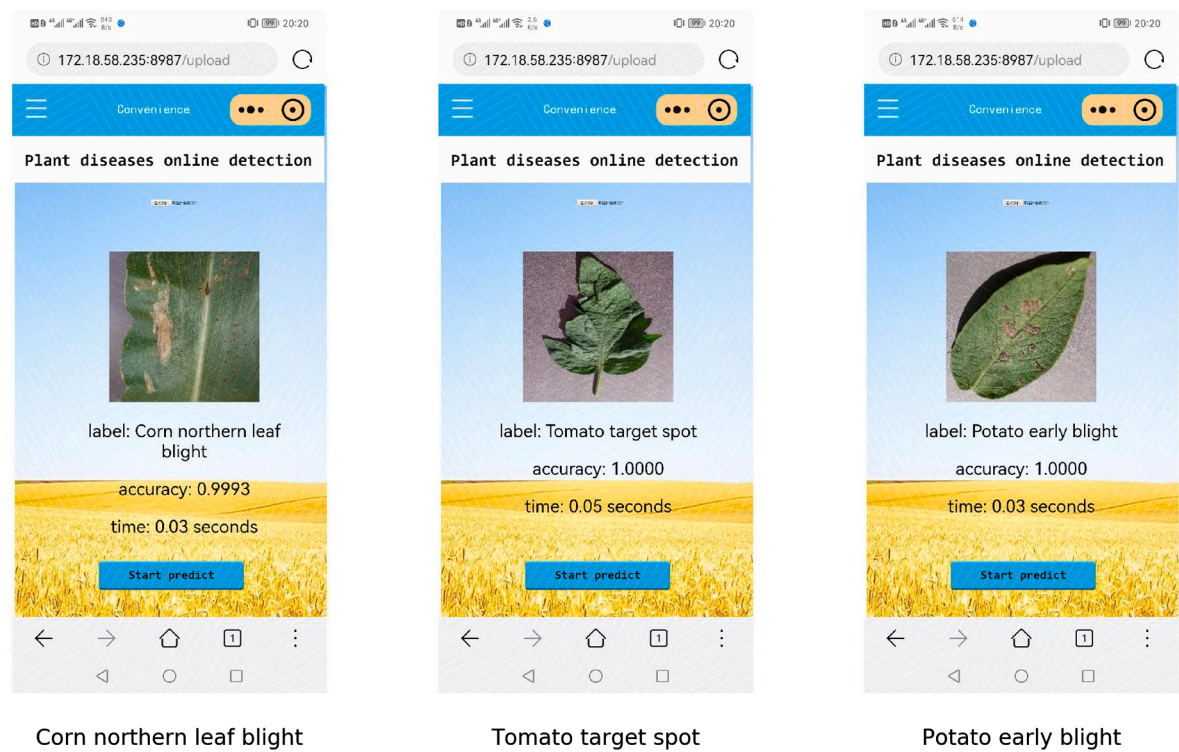


Fig. 12. Mobile phone terminal: web-based real-time disease detection results.

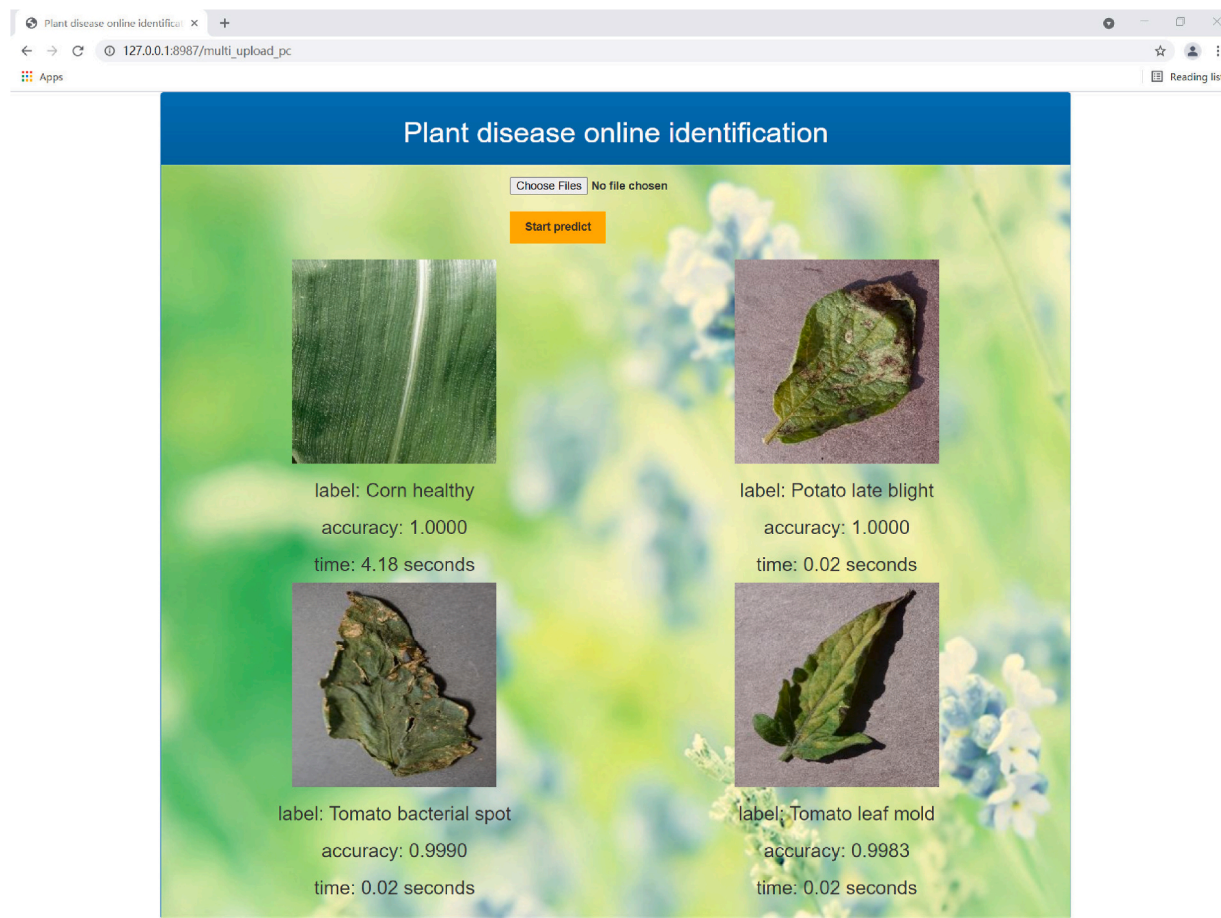


Fig. 13. Computer terminal: web-based real-time disease detection results.

Table 11
Accuracy of classification of multiple diseases.

Network	Dataset	Accuracy
Modified MobileNet (Kamal et al., 2019)	PlantVillage	97.65%
Proposed model (Chohan et al., 2020)	PlantVillage	98.3%
RIC-Net(our model)	PlantVillage(corn, tomato, potato)	99.55%

Table 12
Accuracy of classification of a single disease.

Network	Corn	Potato	Tomato
Optimized DenseNet (Waheed et al., 2020)	98.06%		
Proposed model (Agarwal et al., 2020)			91.2%
RIC-Net(our model)	98.44%	99.43%	95.20%

corn, potato and tomato, respectively. To meet the demand for plant disease detection in real-life scenarios, we also designed and developed a web-based real-time disease detection system. The system can run on both cell phones and computers. In the test, the system showed high performance and could identify plant diseases quickly and accurately. In our future work, we will focus on three key tasks. The first task is to extend the crop disease categories to make our model more generalizable. The second task is to deploy our proposed model on agricultural robots. The third task is to place the agricultural robots in real scenarios

in order to achieve an integrated process of data collection and detection in real environments.

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CRediT authorship contribution statement

Yun Zhao: Conceptualization, Methodology, Software, Writing – review & editing, Supervision, Funding acquisition. **Cheng Sun:** Conceptualization, Methodology, Software, Validation, Resources, Writing – original draft, Writing – review & editing. **Xing Xu:** Investigation, Resources, Data curation, Visualization, Supervision. **Jiagui Chen:** Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

training loss curves of 8 trained models

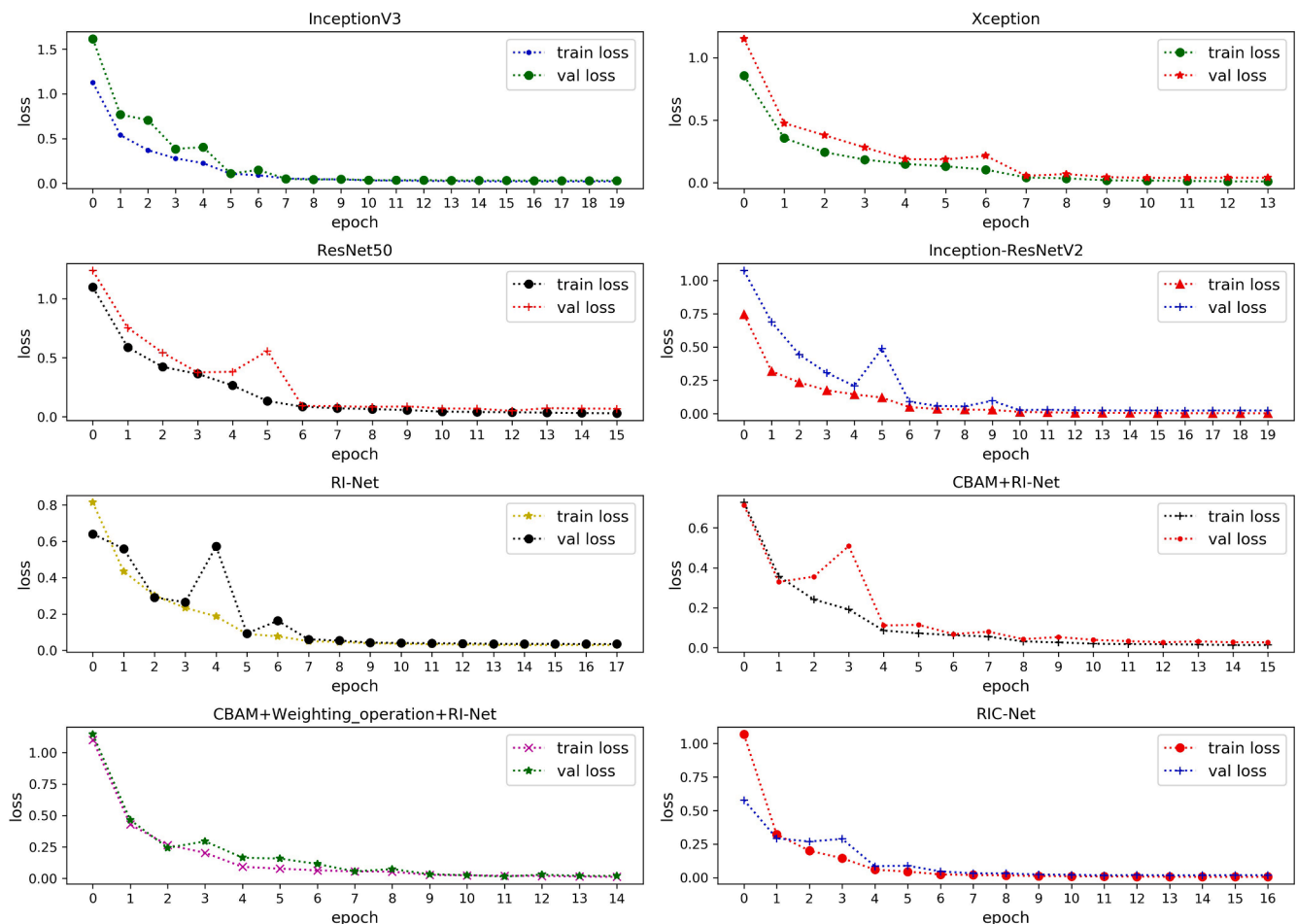


Fig. A1. The training loss curves of 8 trained models.

learning rate curves of 8 trained models

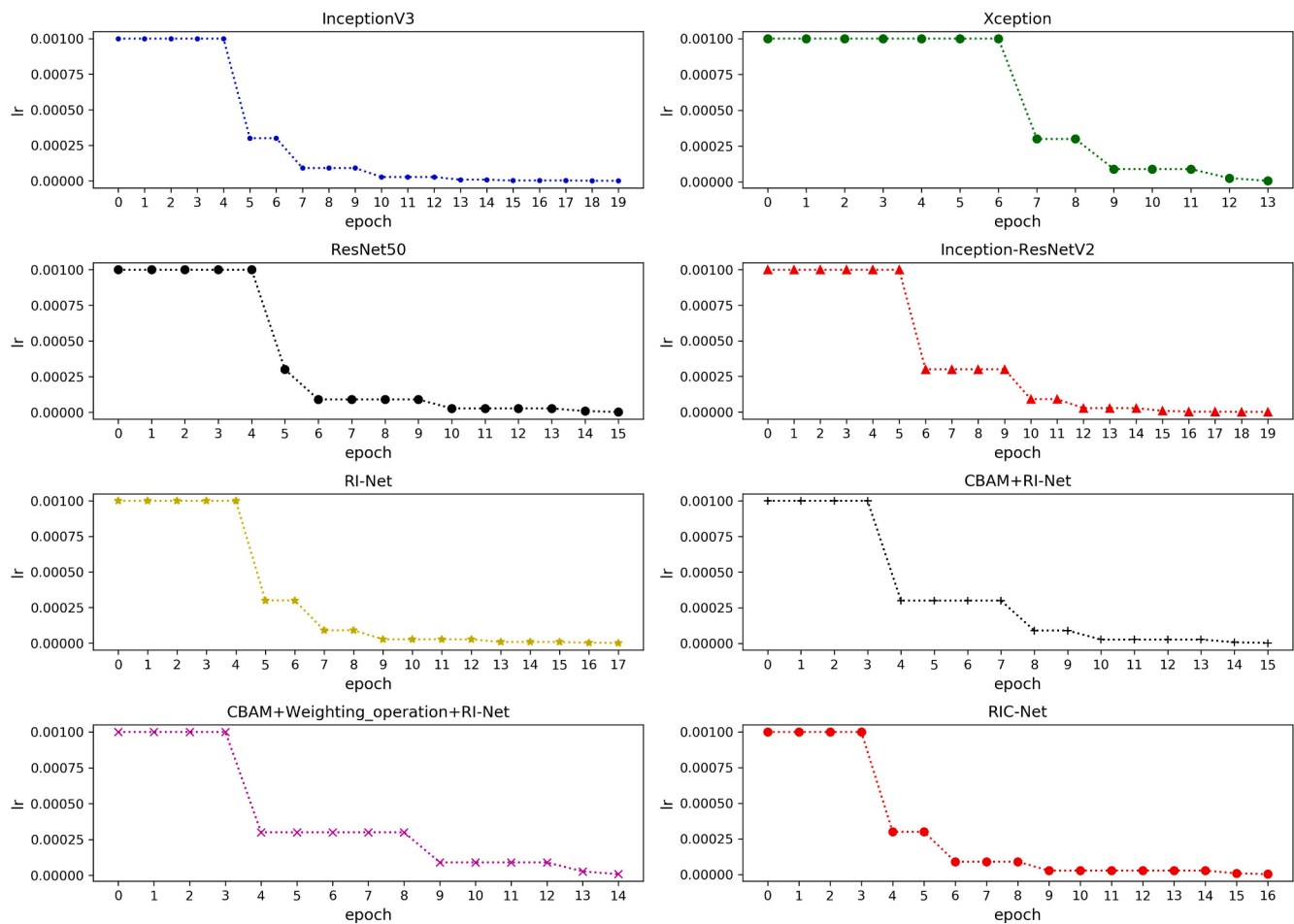


Fig. A2. The learning rate decline curves of 8 trained models.

Table A1

Detailed training results of 8 models.

Model	Training accuracy%	Validation accuracy%	Testing accuracy%
InceptionV3	99.42	99.05	99.08
Xception	99.48	98.51	98.71
ResNet50	98.71	98.36	98.41
Inception-ResNetV2	99.93	99.13	99.08
RI-Net	99.12	99.01	98.91
CBAM + RI-Net	99.48	98.96	99.01
CBAM + Weight_operation + RI-Net	99.27	99.43	99.36
RIC-Net	99.80	99.35	99.33

Appendix A

Fig. A1 and Fig. A2, Table A1.

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